The EuRoC MAV Dataset

This web page presents visual-inertial datasets collected on-board a Micro Aerial Vehicle (MAV). The datasets contain stereo images, synchronized IMU measurements, and accurate motion and structure ground-truth.

Those data sets were published in:

M. Burri, J. Nikolic, P. Gohl, T. Schneider, J. Rehder, S. Omari, M. Achtelik and R. Siegwart, *The EuRoC micro aerial vehicle datasets*, **International Journal of Robotic Research**, DOI: 10.1177/0278364915620033, early 2016. Publisher Link

Available Data

- Visual-Inertial Sensor Unit
  - **Stereo Images** (Aptina MT9V034 global shutter, WVGA monochrome, 2×20 FPS)
  - **MEMS IMU** (ADIS16448, angular rate and acceleration, 200 Hz)
  - Shutter-centric temporal alignment
- Ground-Truth
  - **Vicon** motion capture system (6D pose)
  - **Leica MS50** laser tracker (3D position)
  - Leica MS50 3D structure scan
- Calibration
  - Camera intrinsics
  - Camera-IMU extrinsics
  - Spatio-temporally aligned ground-truth

Downloads

Dataset	Comment
Machine Hall 01	Dataset machine hall “easy”
Machine Hall 02	Dataset machine hall “easy”
Machine Hall 03	Dataset machine hall “medium”
Machine Hall 04	Dataset machine hall “difficult”
Machine Hall 05	Dataset machine hall “difficult”
Vicon Room 1 01	Dataset Vicon room 1 “easy”
Vicon Room 1 02	Dataset Vicon room 1 “medium”
Vicon Room 1 03	Dataset Vicon room 1 “difficult”

Vicon Room 2 01	Dataset Vicon room 2 “easy”
Vicon Room 2 02	Dataset Vicon room 2 “medium”
Vicon Room 2 03	Dataset Vicon room 2 “difficult”
Calibration Dataset	Dataset for custom calibration

Calibration All extrinsic and intrinsic calibration parameters plus post-processed ground-truth are contained in the downloads above (ASL Dataset Format).

Dataset Parser A simple dataset parser is available here: dataset_tools. If you use ROS and you are unable to process bulks of data, you can find a python script that re-assigns the correct header time-stamps to the bag file.

Datasets Two batches of datasets are available. The first batch was recorded in the ETH machine hall (see Figure 1) and contains millimeter accurate position ground-truth from a Leica MS50 laser tracker. The second batch contains Vicon 6D pose ground truth and a precise, registered 3D scan of the environment (see Figure 2).

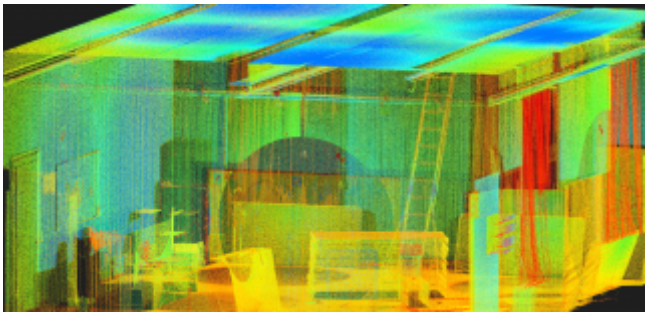

Figure 1: ETH Machine hall. Figure 2: ground-truth 3D scan of the Vicon environment.

Platform and Sensors

An Asctec Firefly hex-rotor helicopter was used for dataset collection, carrying a visual-inertial (camera-IMU) sensor unit (see Figure 3).

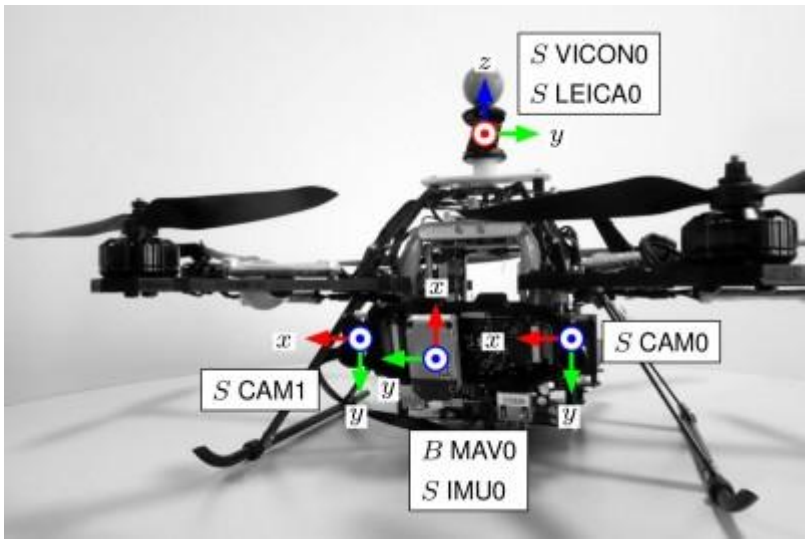

Figure 3: Asctec Firefly hex-rotor helicopter used during dataset collection.

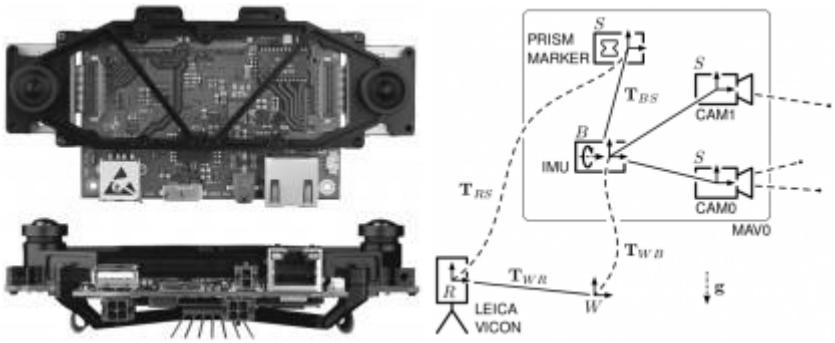

Figure 4: Left: Visual-Inertial sensor unit (carried by the helicopter). Right: Sensors and ground-truth instruments schematic overview.

Ground-Truth

Instruments

The following devices were used to capture 6D motion and structure ground-truth

1. Leica MS50 laser tracker and scanner
2. Vicon 6D motion capture system.

Spatio-Temporal Alignment

The sensor and ground-truth data are extrinsically calibrated, and temporally aligned.

Known Issues

- The visual-inertial sensor employs an automatic exposure control that is independent for both cameras. This resulted in different shutter times and in turn in different image brightnesses, rendering stereo matching and feature tracking more challenging. Since the mid-exposure times of both cameras were temporally aligned, synchronization was not affected by different shutter times.
- Some of the datasets exhibit very dynamic motions, which are known to deteriorate the measurement accuracy of the laser tracking device. The numbers reported by the manufacturer may be overly optimistic for these events, which complicates the interpretation of ground truth comparisons for highly accurate visual odometry approaches. The effect on sections with less dynamic motion—particularly the start and end of each dataset—is assumed to be negligible.
- The accuracy of the synchronization between sensor data and motion ground truth is limited by the fact that both sources were recorded on different systems and that device timestamps were unavailable for the Vicon system. This issue is mitigated by estimating the temporal offset Δt S as a state, addressing both a fixed temporal offset and clock drift. Where available, device timestamps were employed to avoid jitter. Raw data is available for assessing alternative synchronization schemes.